Lecture 20:

VLSI High-Speed I/O Transmitters Circuits

Contents

This Topic: VLSI High-Speed I/O Transmitter Circuits

- Introduction
- Basic Tx Blocks and Design Aspects
- Timing Aspect of High-Speed Tx Design
- VLSI Tx Circuit Implementations
- PAM-4 Transmitter

Introduction

- Tx offers necessary signal conditioning to ensure transmission quality at high data rate
- Key Tx design challenges are to minimize datapath delay variation under wide application conditions
  - PVT variations
  - Loading (Cap loads, T-Line/Trace discontinuities)
- Tx must meet certain key design specifications
  - Signal Swing
  - Impedance matching (Termination, Cpad, S11 etc.)
  - Equalization
  - Slew Rate

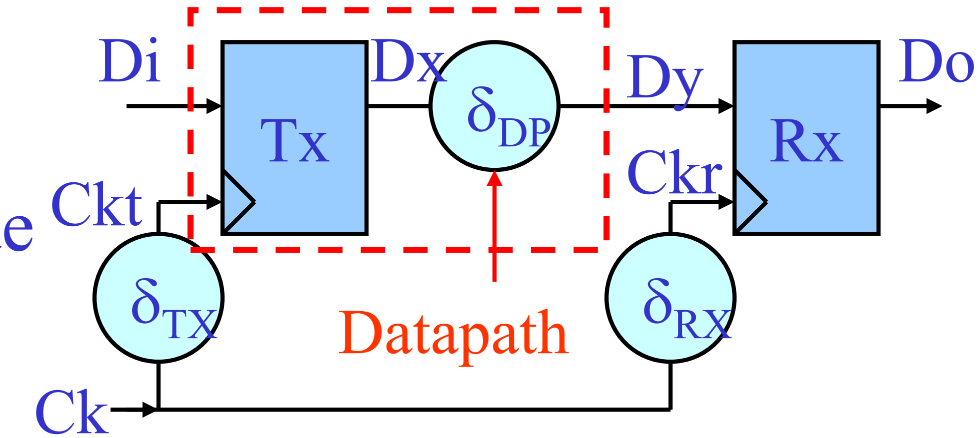
Introduction: VLSI Tx Design Considerations

- Tx design considerations
  - Ringing/Reflections
  - PVT variations
  - Power Supply Noise
  - X-talk
  - ISI
  - Clock reference
  - ESD
- Tx should provide high quality signal across wide load conditions!!

Clock skews

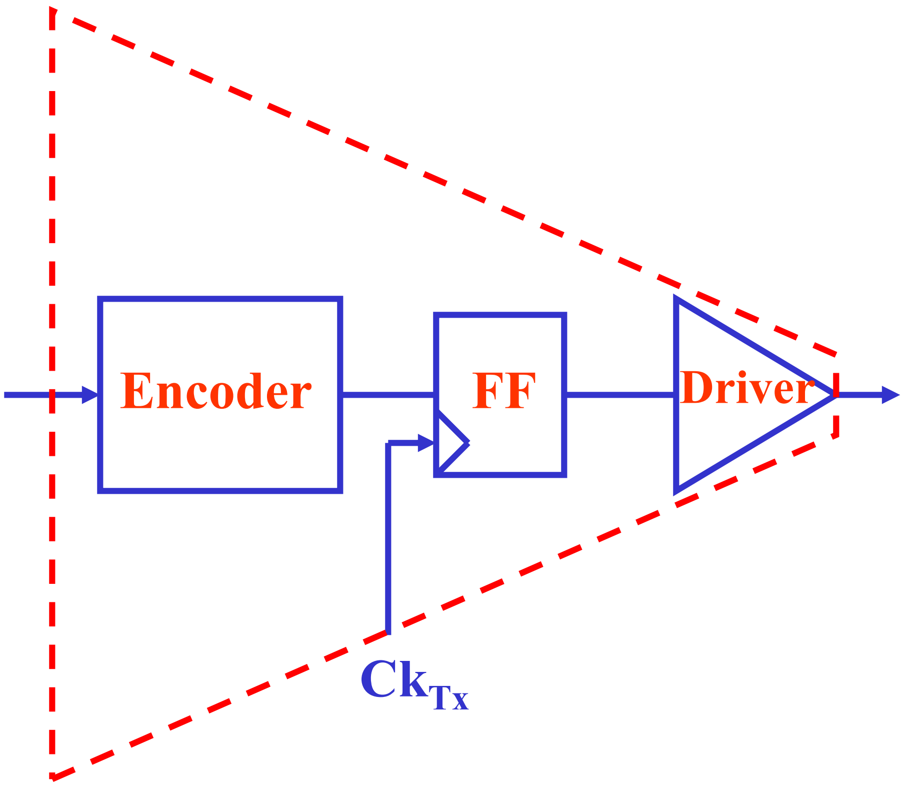
$$t_{hold} < \delta_{DP} + \delta_{Tx} - \delta_{Rx} < T - t_{setup}$$

Datapath Delay



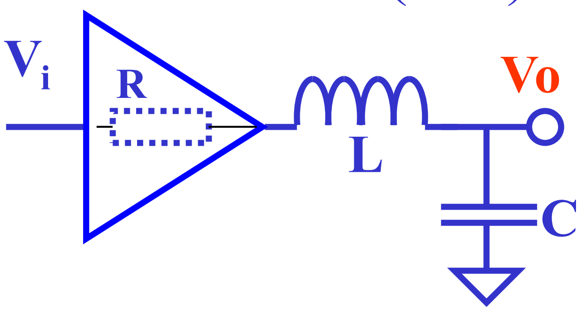
Basic High-speed I/O Tx Circuit Components

- Tx Driver
  - Loading
  - ESD
  - BW
  - Signal integrity (SI)
- Sync Element (FF)
  - Timing
- Encoder
  - Channel Coding
  - Scrambling
  - PISO, etc.

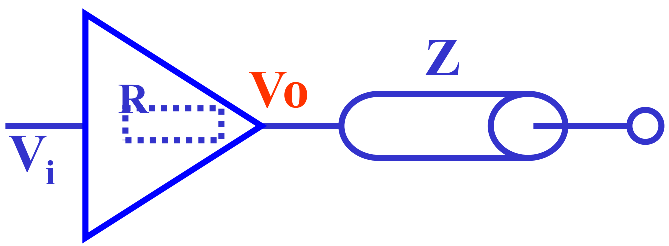


Tx Driver Design: Tx R-Term for Ringing/Reflection Control

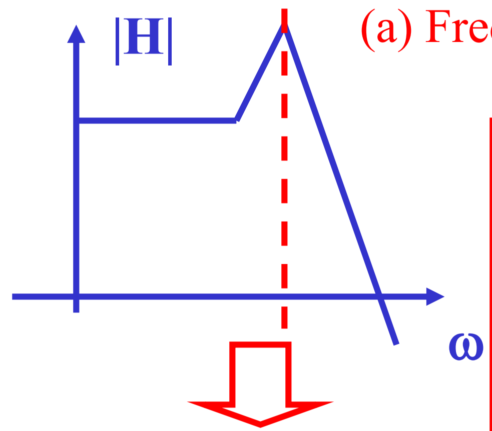
- Critical damping is the key to eliminate ring/reflection
  - (Q = 0.7~1).
  - Tx R-term to eliminate reflection (R = Z)
  - $R = Z = (L/C)^{0.5}$



(a) Lumped model



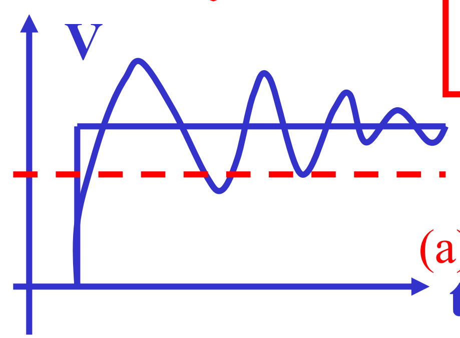
(b) Distributed model



(a) Frequency-domain model

$$\frac{V_o}{V_i} = \frac{1}{(\frac{s}{\omega_o})^2 + \frac{1}{Q}(\frac{s}{\omega_o}) + 1}$$

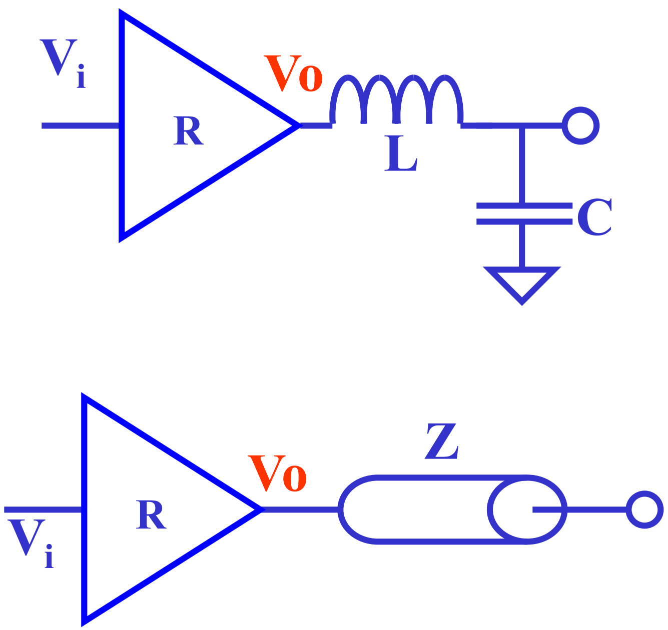
$$\omega_o = \frac{1}{\sqrt{LC}}, Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

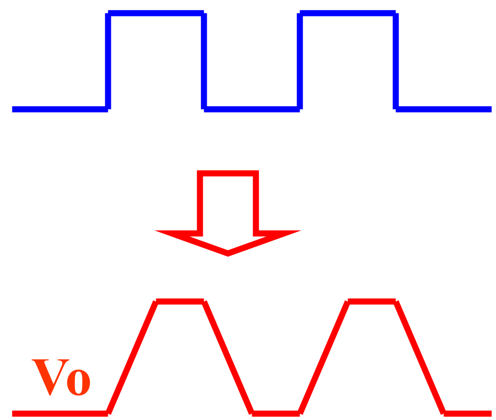


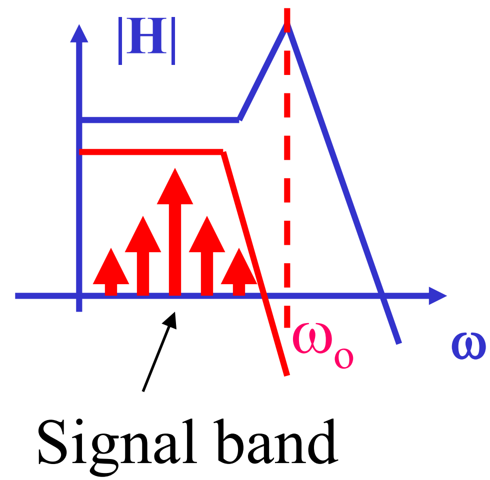
(a) Time-domain model

Tx Driver Design: Tx Slew Rate for Ringing/Reflection Control

- Slew Rate control to minimize ringing/reflection



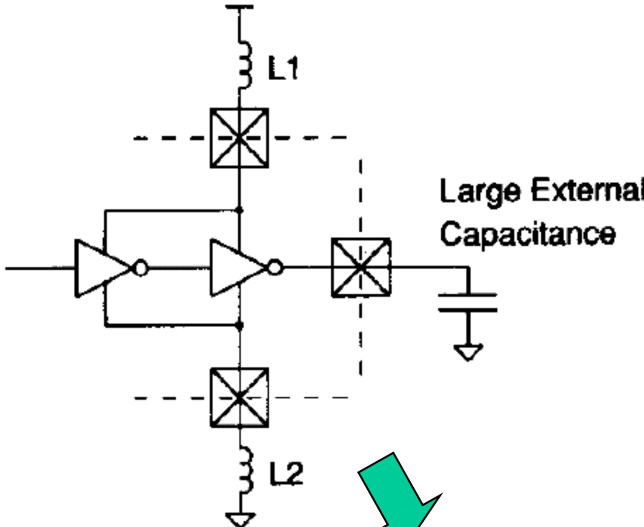


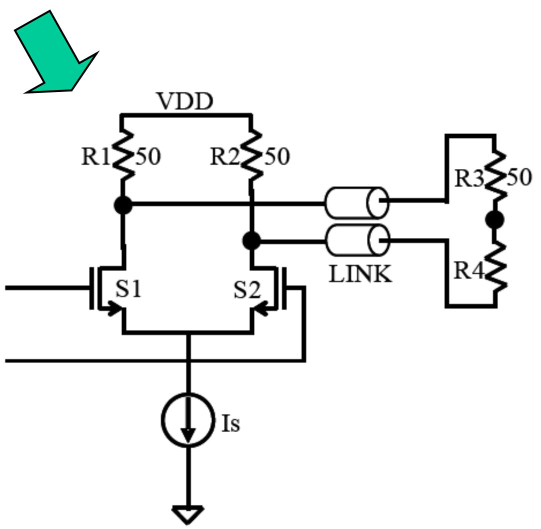


Not very effective if ω_o is low!!

Tx Driver Design: Supply Noise Control

- Minimization of switching noise
  - Fully differential driver
  - Low swing driver
  - Low inductor supply
  - Low switching current
  - Use of decoupling Cap
  - Slew Rate control





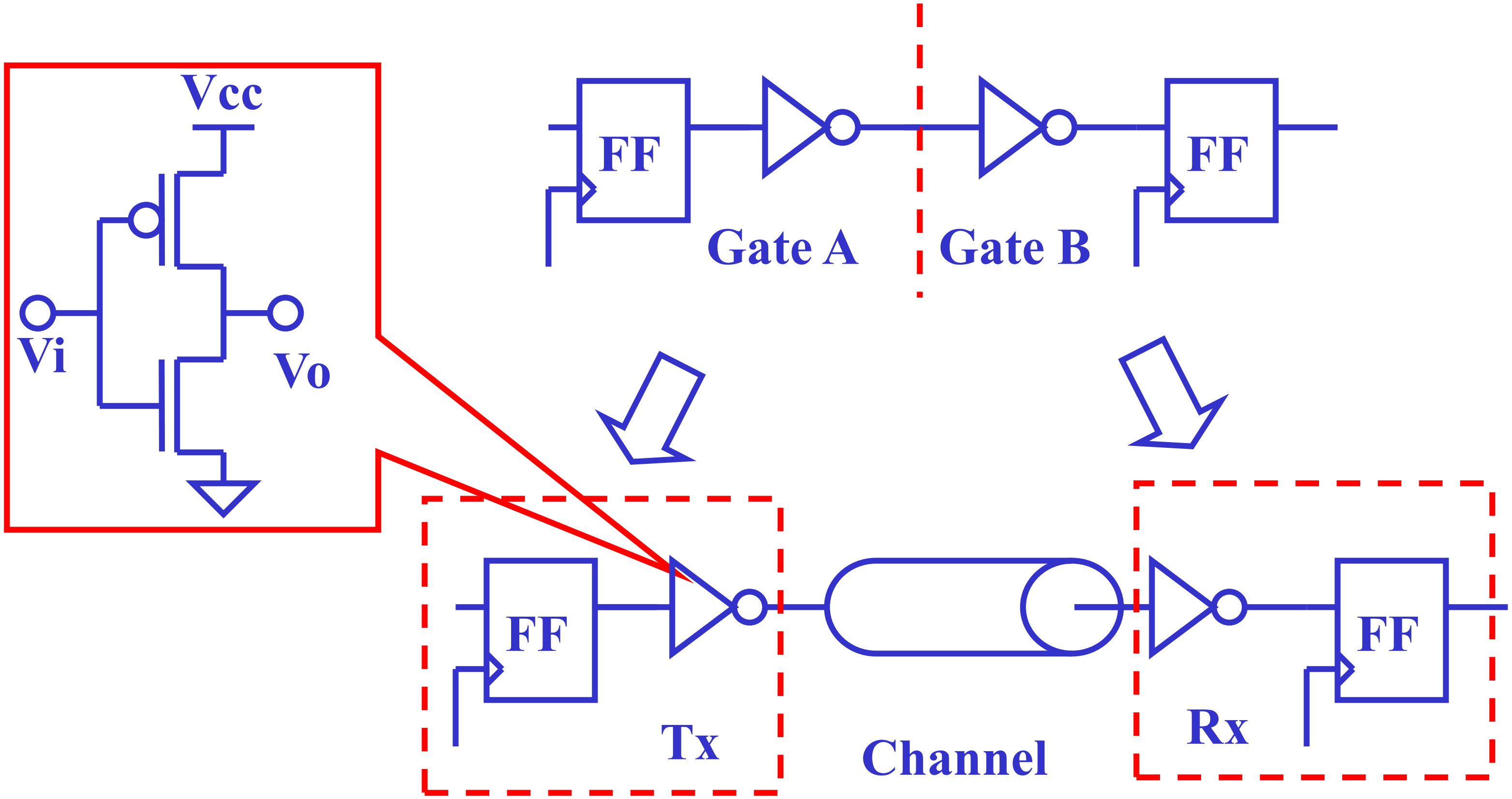
- Minimization of IR drop
  - Low impedance supply

$$\boxed{\Delta V_{DD} = IR}$$

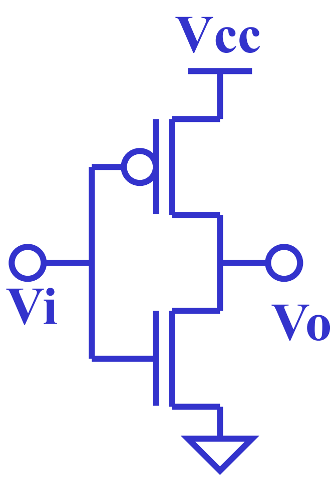
$$\boxed{\Delta V_{DD} = L\frac{dI}{dt}}$$

Traditional Tx Driver: CMOS Inverter Driver

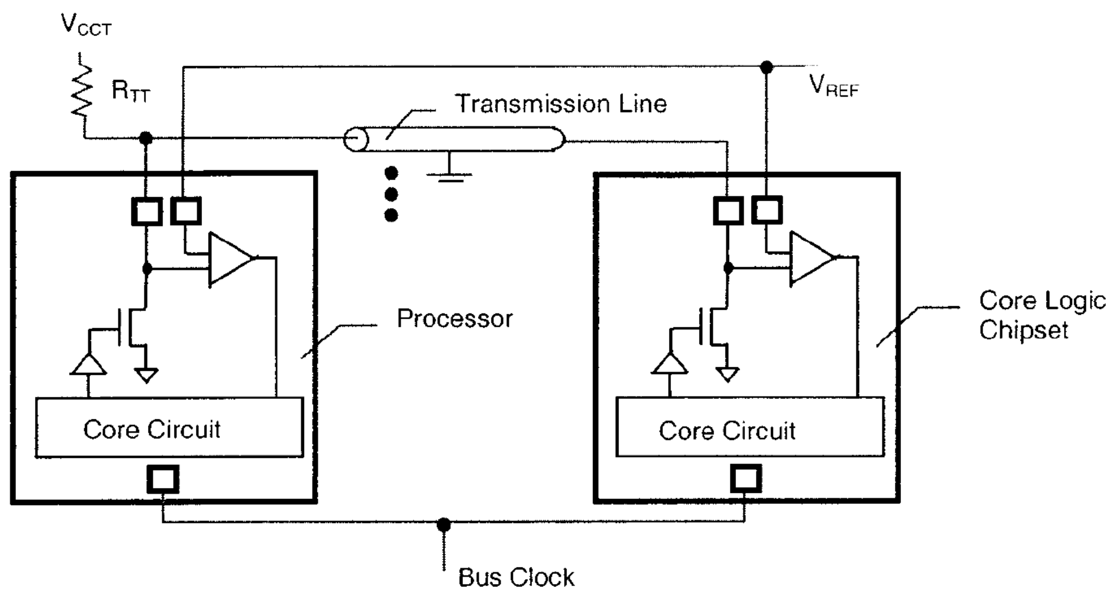
- CMOS driver is a "natural" transition from on-chip to I/O circuits in CMOS technologies



Traditional Tx Driver Circuits



(a) CMOS driver



(b) GTL driver

Characteristics of Traditional Tx Drivers

- Simple design
- High/Full output swings
  - High power dissipation
  - High dI/dt Noise
- Strong PVT dependency
- Strong load dependency
- Poorly controlled slew rate
- Poorly controlled output impedance

High-Speed I/O Driver Circuit Techniques

- Low Swing Signaling
- Differential Signaling

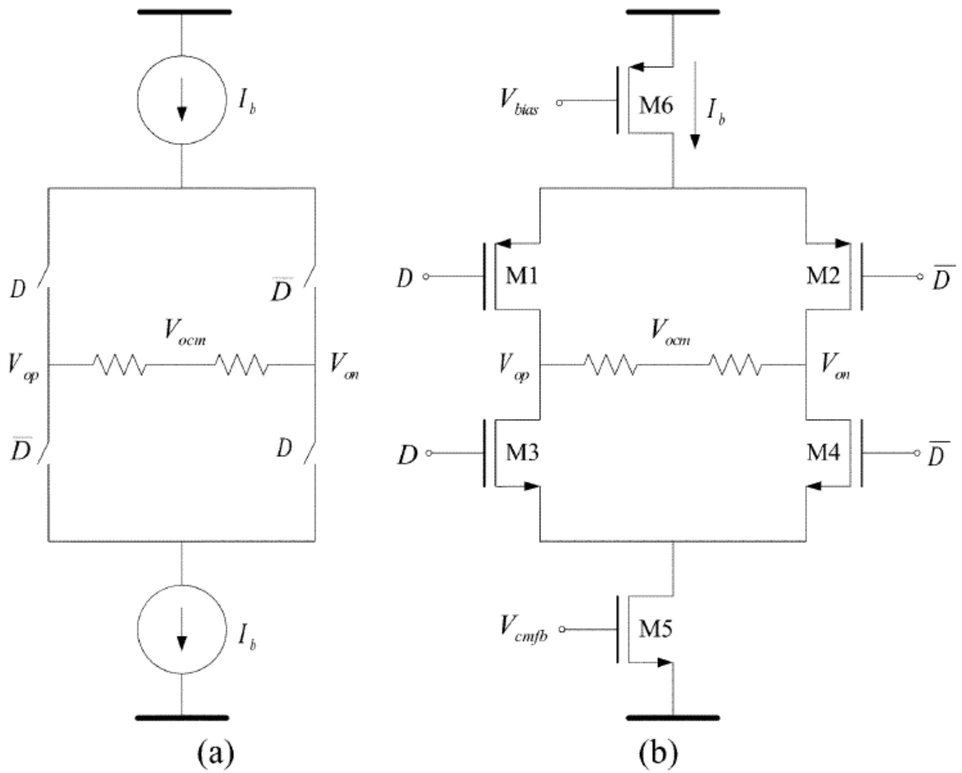
LVDS

- Constant Current Driving
- Improved Voltage mode driver
- Reduced Switching Noise driver
- Impedance Control (termination)
- Slew Rate Control
- Spectra-Shaping (Channel Coding+ Equalization)
- Other Driver Circuits (Linear, Bi-Direction, Multi-level etc.)

Differential Tx Driver Circuits

- Differential drivers offers high noise immunity by eliminating common-mode influences such as
  - Cross Talk
  - Simultaneous switching
  - Power supply
  - Ground Bounce noise
- Differential drivers enables low signal swing for
  - High-Speed
  - Lower Power
  - Lower Noise
  - Lower EMI

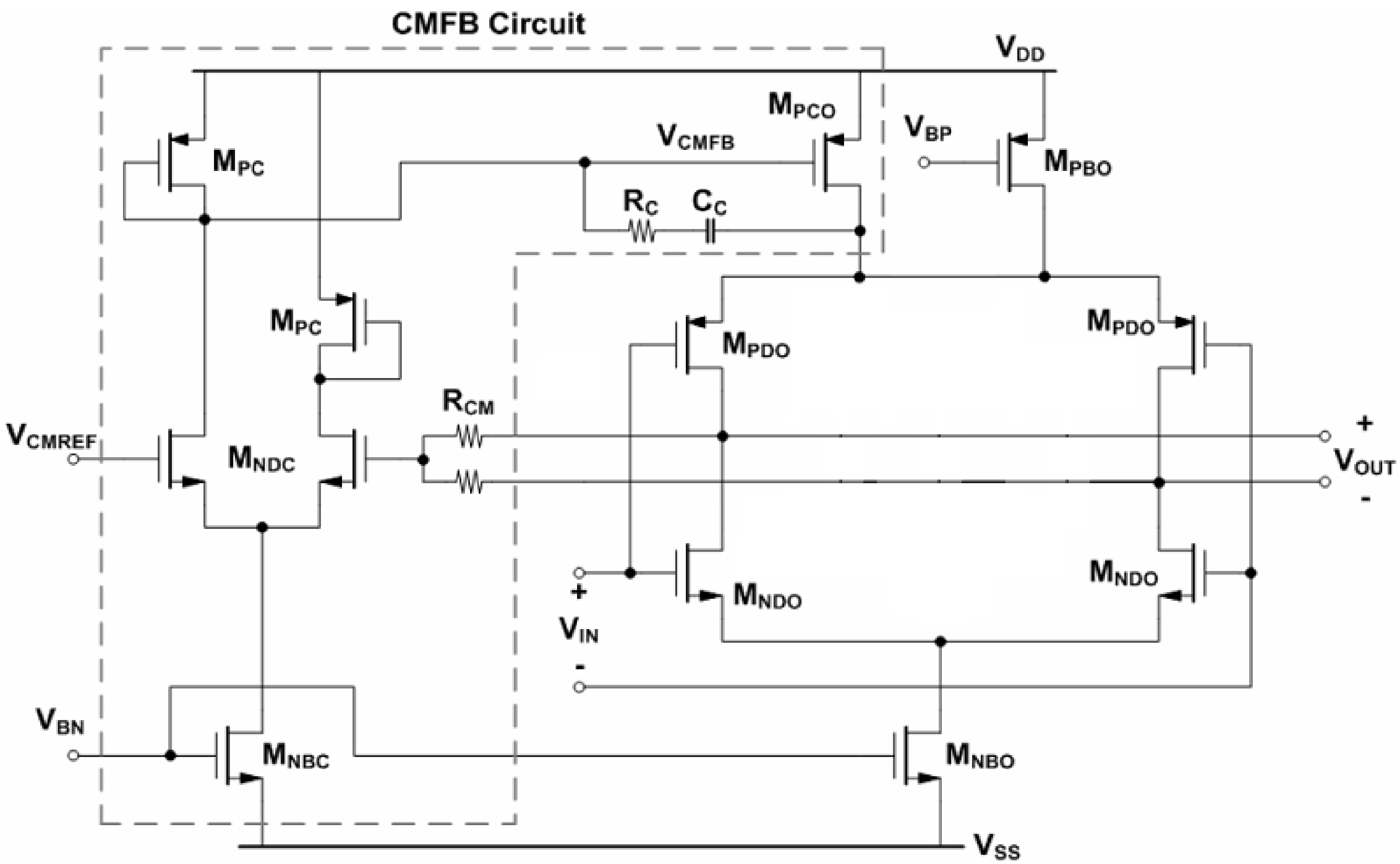
VLSI Differential Tx Driver (I): Push-Pull



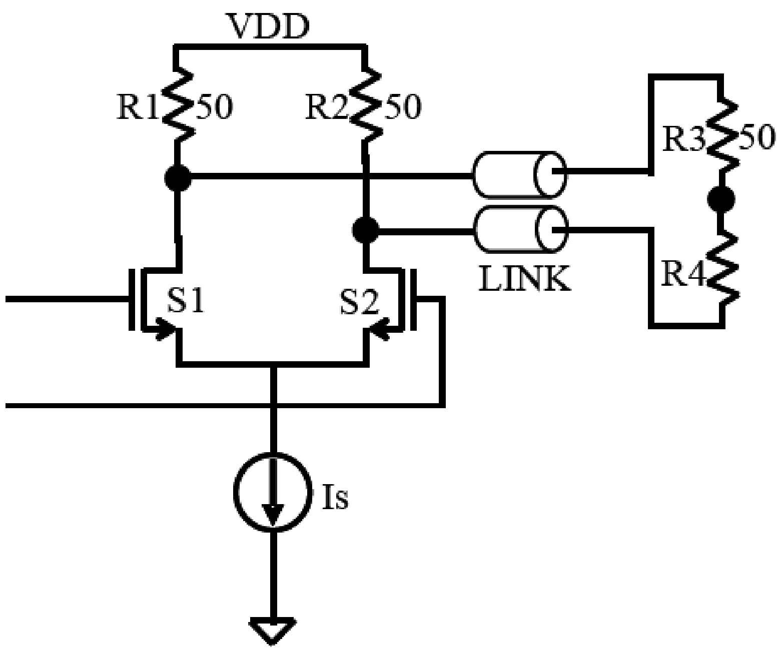
- Push-Pull Driver
- Internal R-Term
- LVDS scheme
- Low dI/dt Noise
- Better PSRR

Fig. 3. Typical LVDS driver: (a) macromodel and (b) transistor implementation [3].

VLSI Push-Pull Tx Circuit Implementations

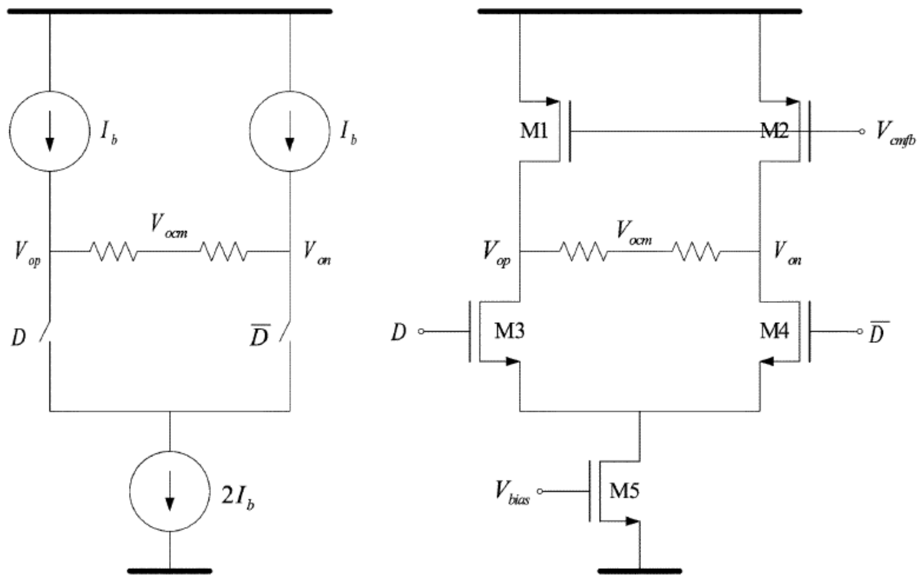


VLSI Differential Tx Driver (II): CML

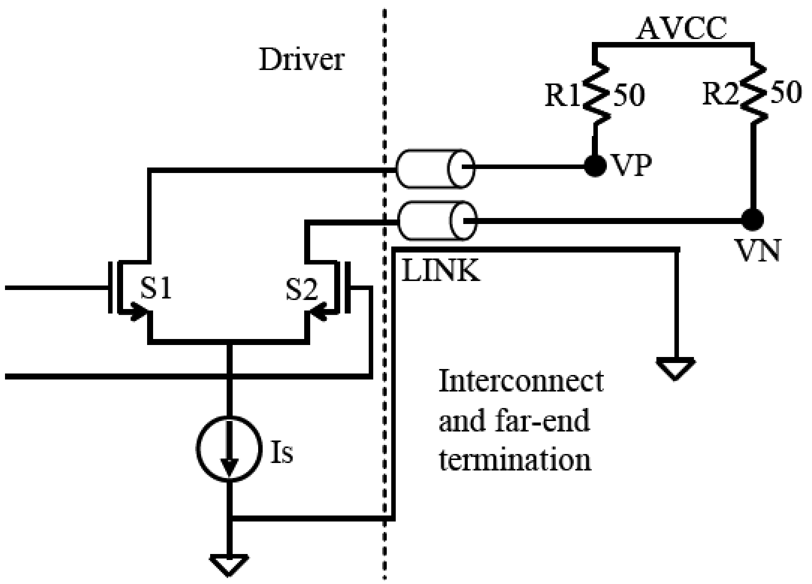


- CML Driver
- Internal R-Term
- LVDS scheme
- Low dI/dt Noise
- Better PSRR
- Lower operation voltage

Alternative VLSI Differential Tx Drivers



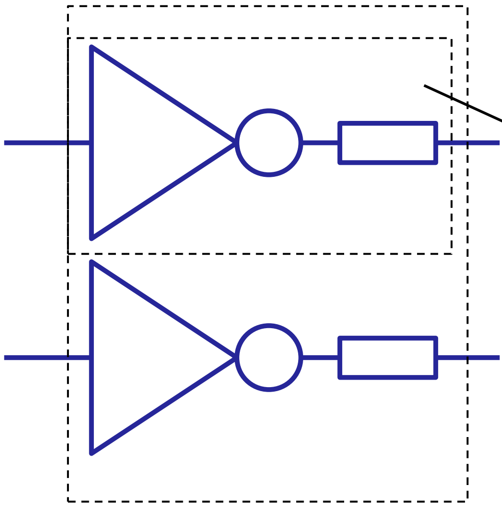
(a)



(b)

VLSI Voltage Mode Driver

- V-Mode driver with programmable impedance



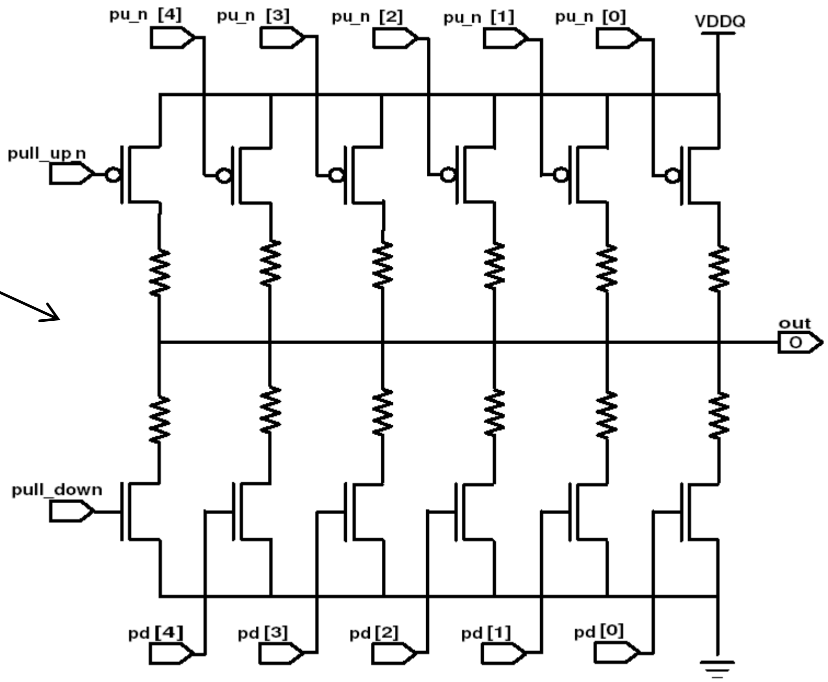
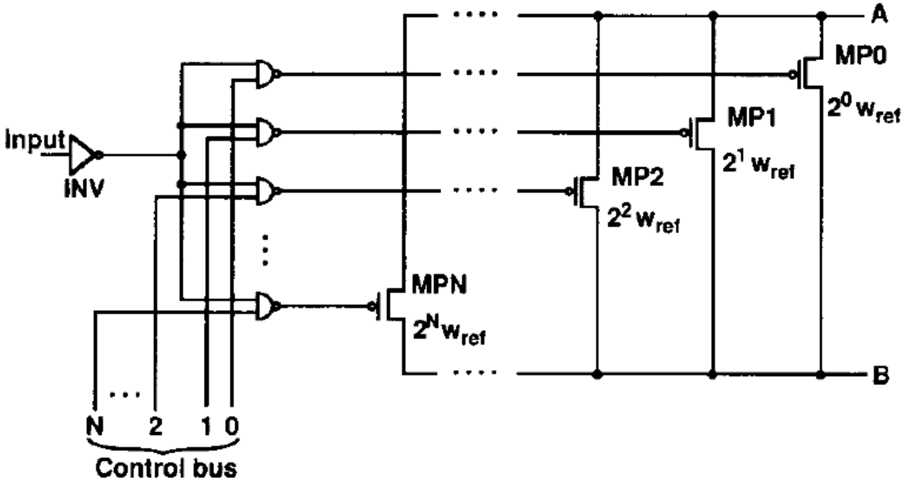


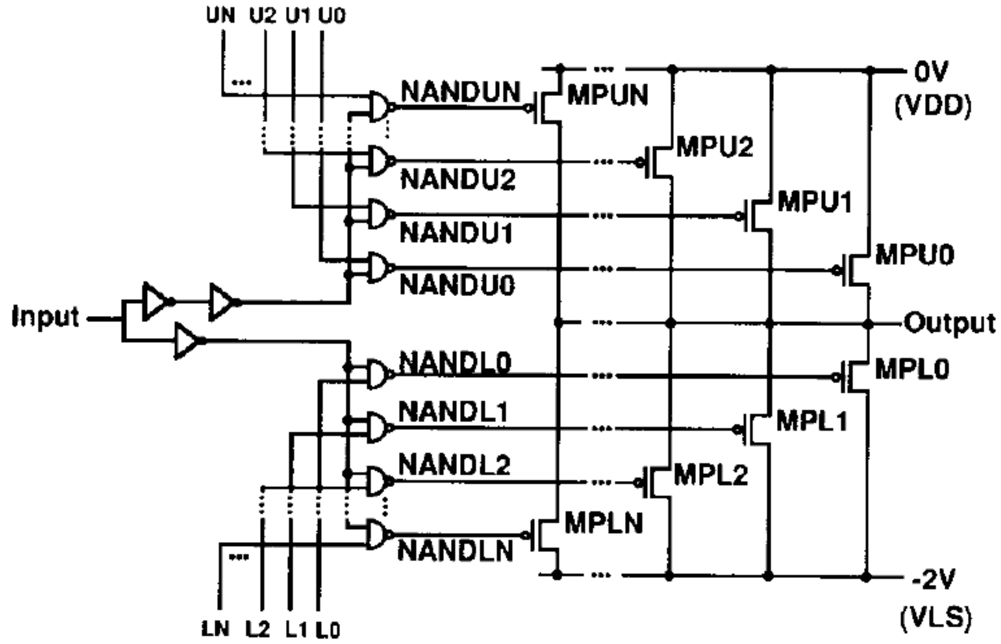
Fig. 5. Poly resistance I/O driver

[Esch04]

Tx Driver Design: Impedance Control

- Programmable Tx output impedance
  - Binary weighted devices





[Gabara92]

Tx Driver Design: Impedance Control

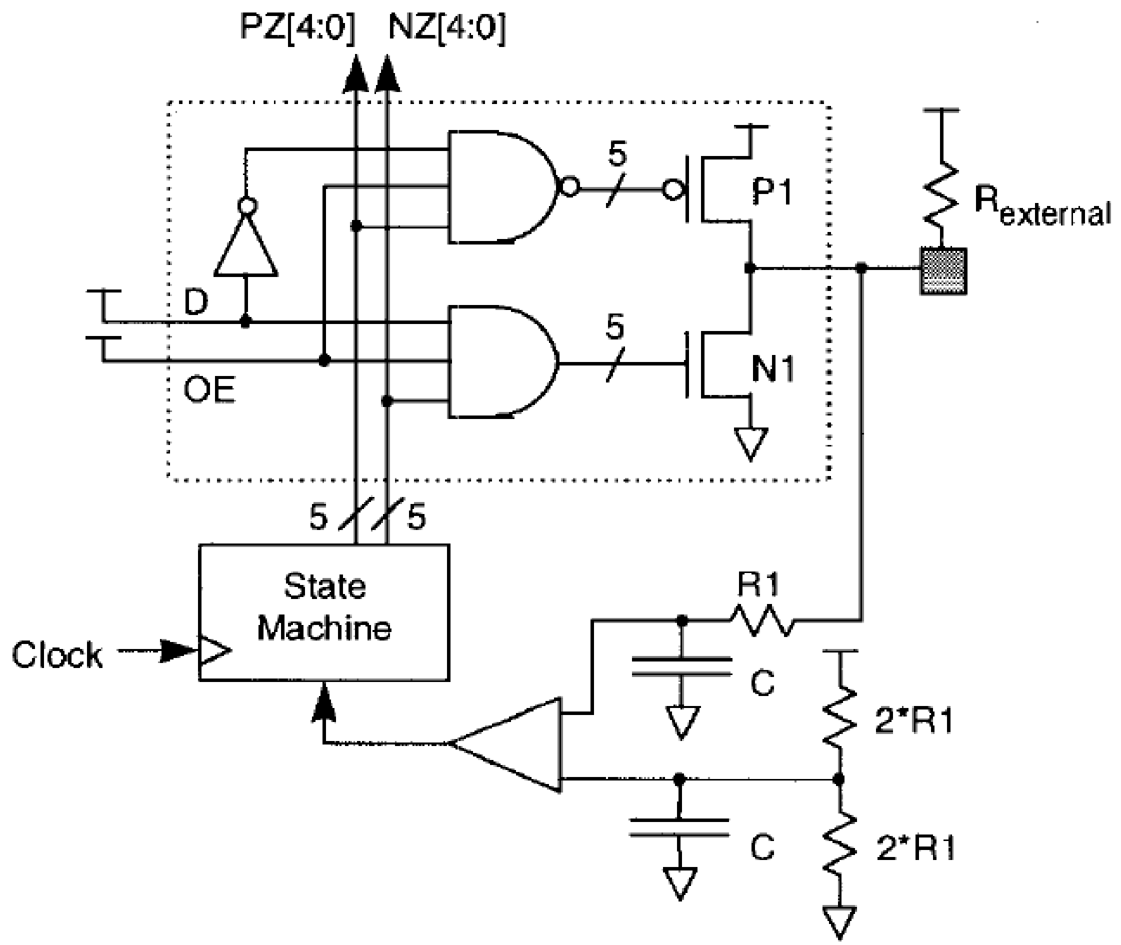


Fig. 3. Impedance control loop

Tx Driver Design: Slew Rate Control

- The effects of inductive overshoot or undershoot of data signal raises serious problem to I/O timing margin
- Effects of Slew Rate Control
  - Reduced Ringing/Reflection
  - Reduced Cross-Coupling

Tx Driver Design: Slew Rate Control

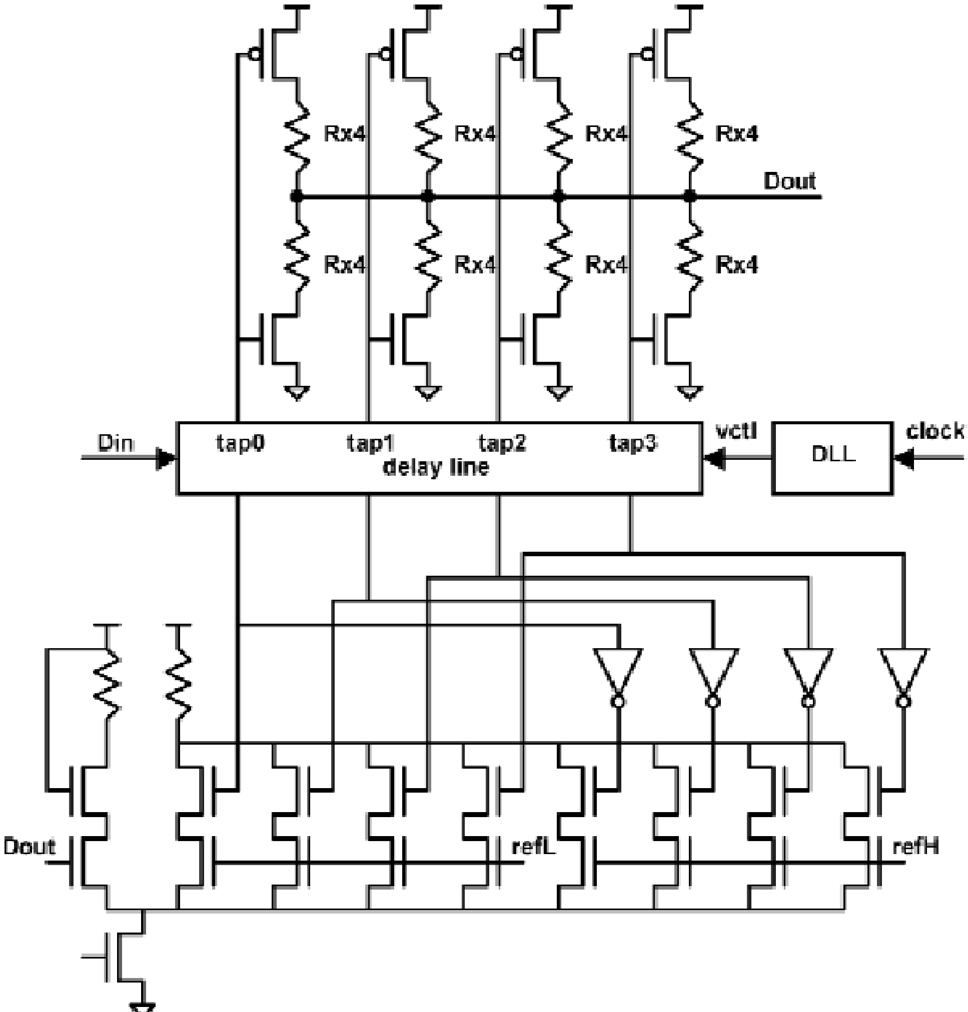
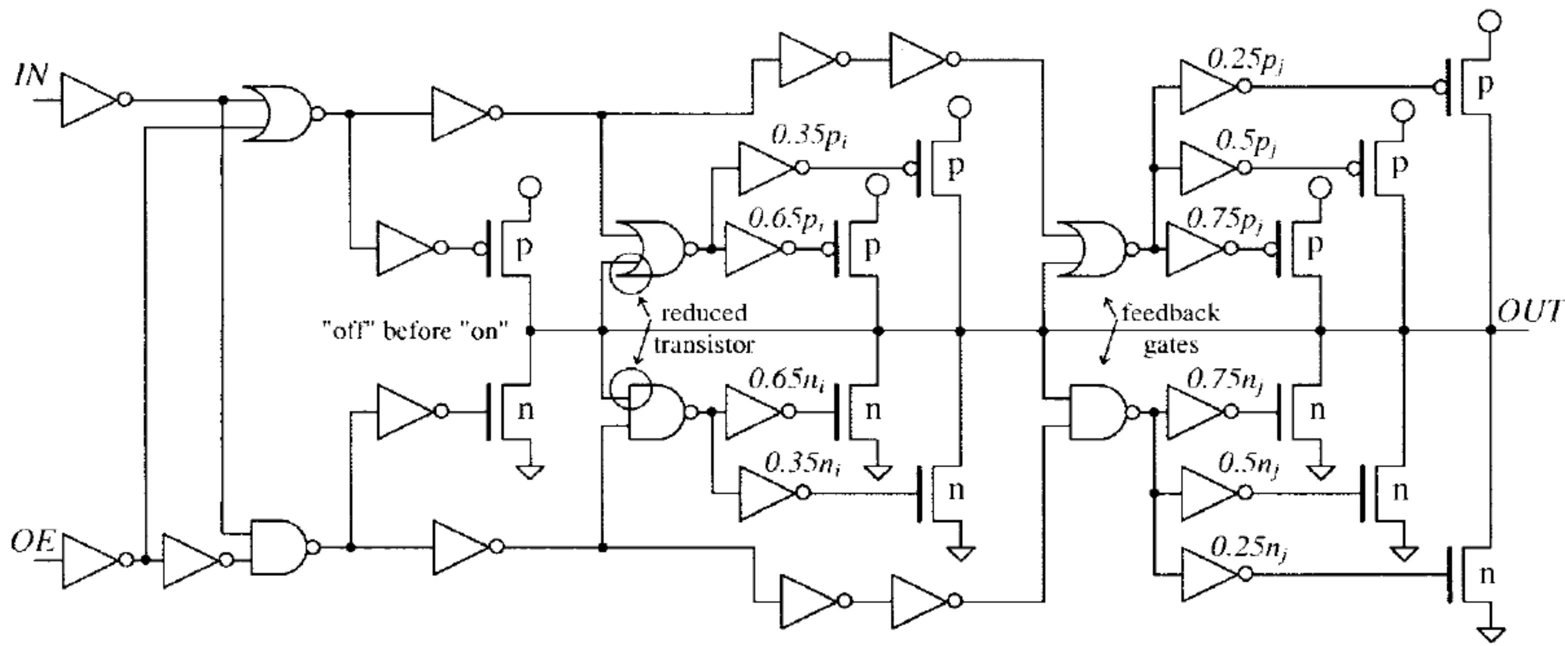


Figure 4.3.2: Controlled slew rate I/O cell.

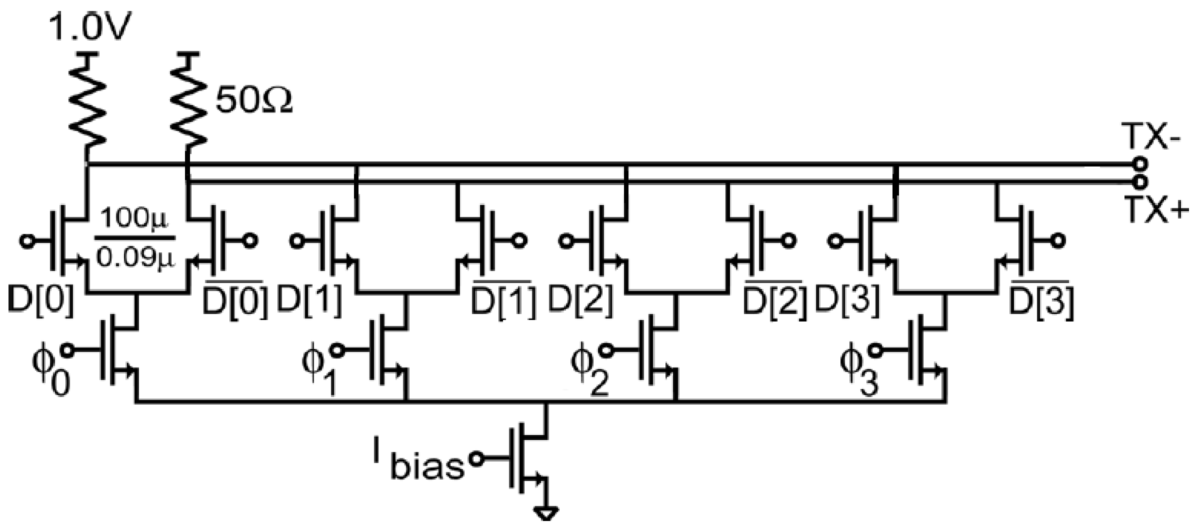
[Haycock01]

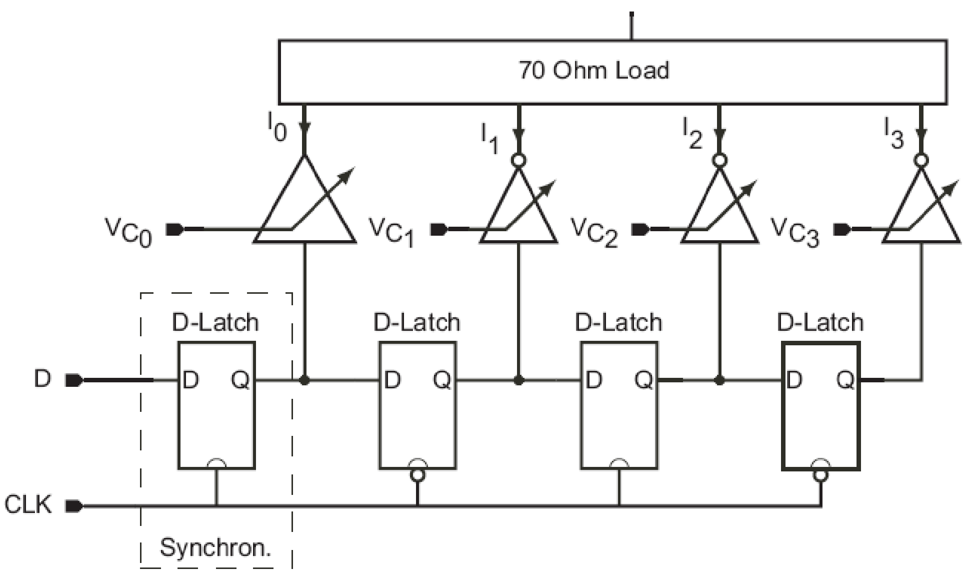
Tx Driver Design: Slew Rate Control



Circuit of adaptive controlled slew-rate output driver (ACSR6).

Tx Driver Design: Equalization





Tx Driver Design: Equalization

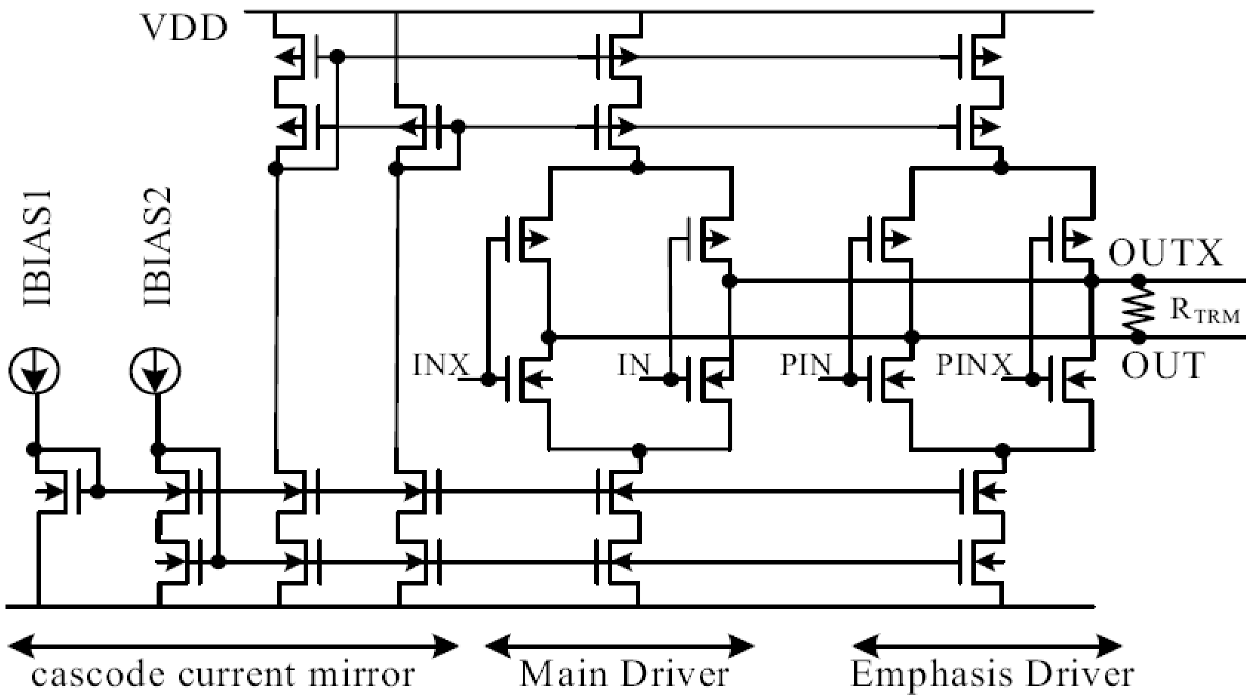


Fig. 4 LVDS driver

Tx Encoder

- Coding for Desired Properties
  - DC Balance, Low disparity
  - Bounded run length
  - High Coding Efficiency
  - Spectral Properties (Low BW and/or DC component)
- Commonly used Coding
  - Manchester
  - 3B4B, 4B5B, 8B10B
  - Scrambling
  - CIMT, Conservative Code

3B/4B Code

- 6 Maximum Runlength
- ¾ Coding Efficiency
- Sending Sync Sequence:
  - SyncA (even), SyncA(odd), SyncB(even), SyncB(odd)
- Allows the unambiguous alignment of 4-bit frame

3B Input	4B output	
	Even Words	Odd Words
000	0011	
001	0101	
010	0110	
011	1001	
100	1010	
101	1100	
110	0110	1011
111	0010	1101
SyncA	0111	1110
SyncB	1000	0001

Example: 5B/6B and 3B/4B Codes

5b/6b code

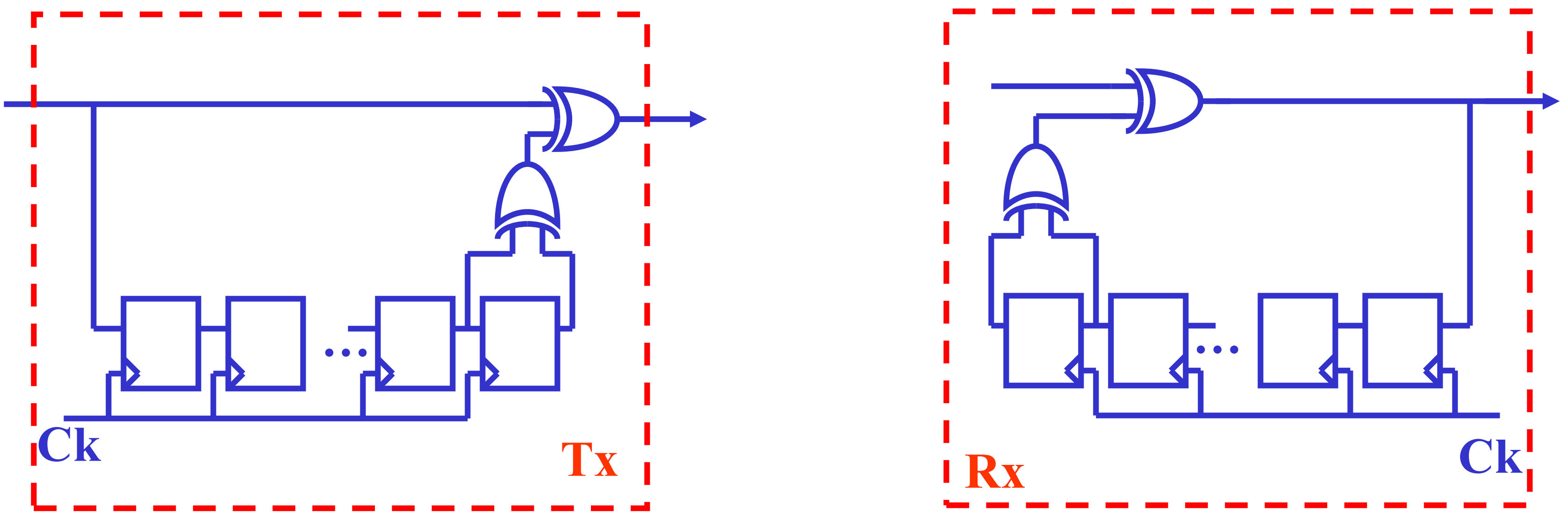
Input		RD = −1	RD = +1	Input		RD = −1	RD = +1
	EDCBA	abcdei			EDCBA	abcdei	
D.00	00000	100111	011000	D.16	10000	011011	100100
D.01	00001	011101	100010	D.17	10001	100011	
D.02	00010	101101	010010	D.18	10010	010011	
D.03	00011	110001		D.19	10011	110010	
D.04	00100	110101	001010	D.20	10100	001011	
D.05	00101	101001		D.21	10101	101010	
D.06	00110	011001		D.22	10110	011010	
D.07	00111	111000	000111	D.23 †	10111	111010	000101
D.08	01000	111001	000110	D.24	11000	110011	001100
D.09	01001	100101		D.25	11001	100110	
D.10	01010	010101		D.26	11010	010110	
D.11	01011	110100		D.27 †	11011	110110	001001
D.12	01100	001101		D.28	11100	001110	
D.13	01101	101100		D.29 †	11101	101110	010001
D.14	01110	011100		D.30 †	11110	011110	100001
D.15	01111	010111	101000	D.31	11111	101011	010100
				K.28	11100	001111	110000

3b/4b code

Input		RD = −1	RD = +1	Input		RD = −1	RD = +1
	HGF	fghj			HGF	fghj	
D.x.0	000	1011	0100	K.x.0	000	1011	0100
D.x.1	001	1001		K.x.1 ‡	001	0110	1001
D.x.2	010	0101		K.x.2 ‡	010	1010	0101
D.x.3	011	1100	0011	K.x.3 ‡	011	1100	0011
D.x.4	100	1101	0010	K.x.4	100	1101	0010
D.x.5	101	1010		K.x.5 ‡	101	0101	1010
D.x.6	110	0110		K.x.6 ‡	110	1001	0110
D.x.P7 †	111	1110	0001				
D.x.A7 †	111	0111	1000	K.x.7 †	111	0111	1000

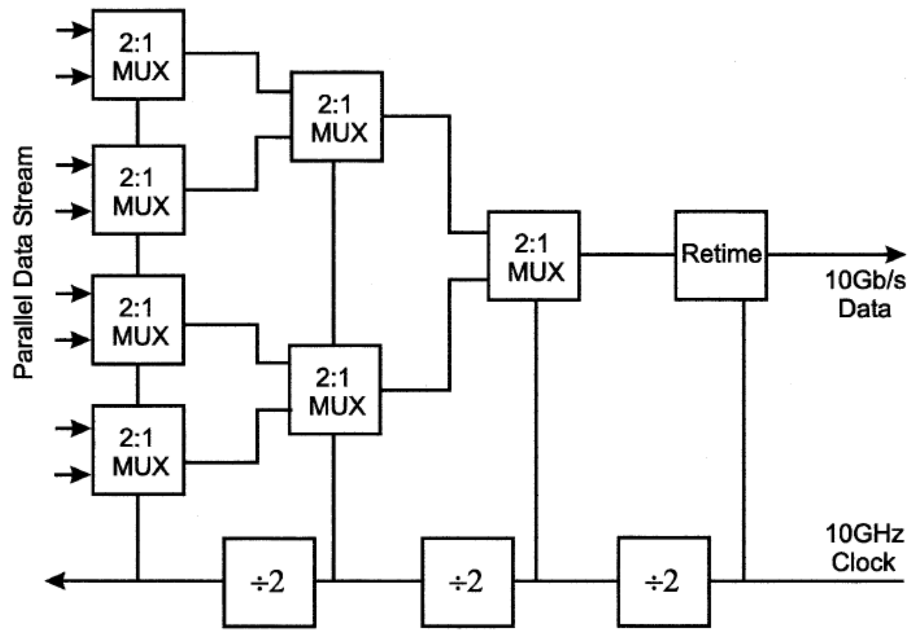
Scrambling

- Use a feedback shift register to randomize data
- De-scrambling (Reversing process) at Rx to restore original data

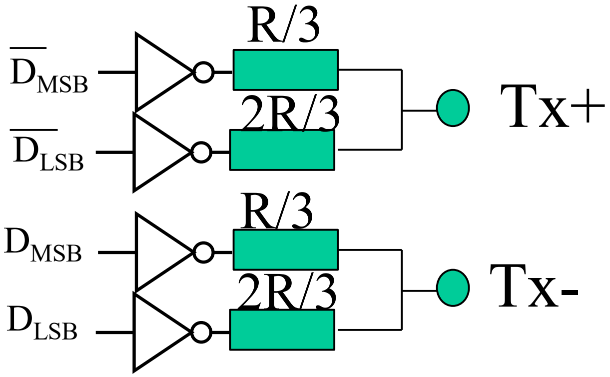


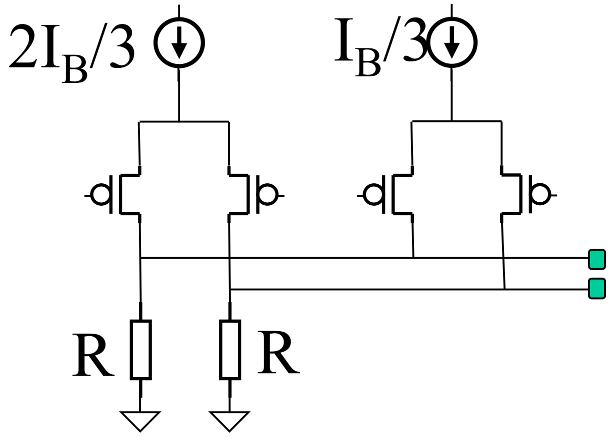
PISO

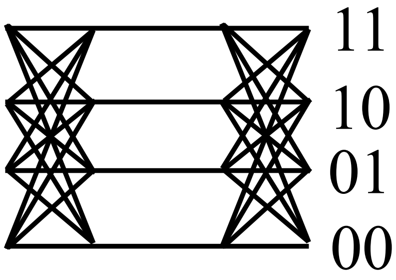
- Parallel input serial output (PISO) is usually used to convert low frequency parallel bit to high frequency serial bit stream for transmission



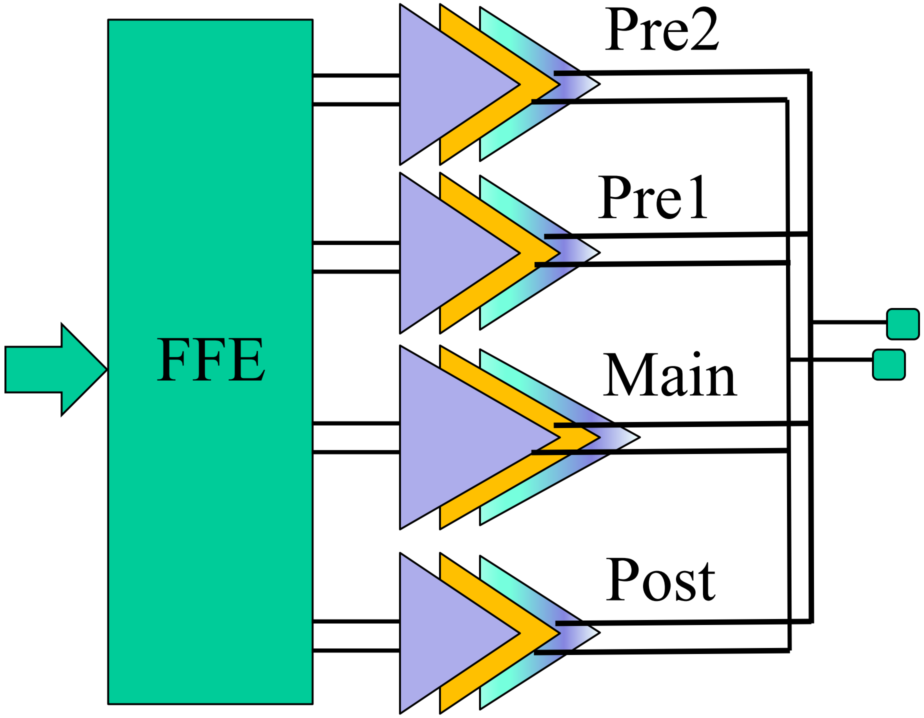
Basic PAM-4 Transmitters







PAM-4 Transmitters+FFE



Reading Assignment

- Please read Chapter 15 of the textbook